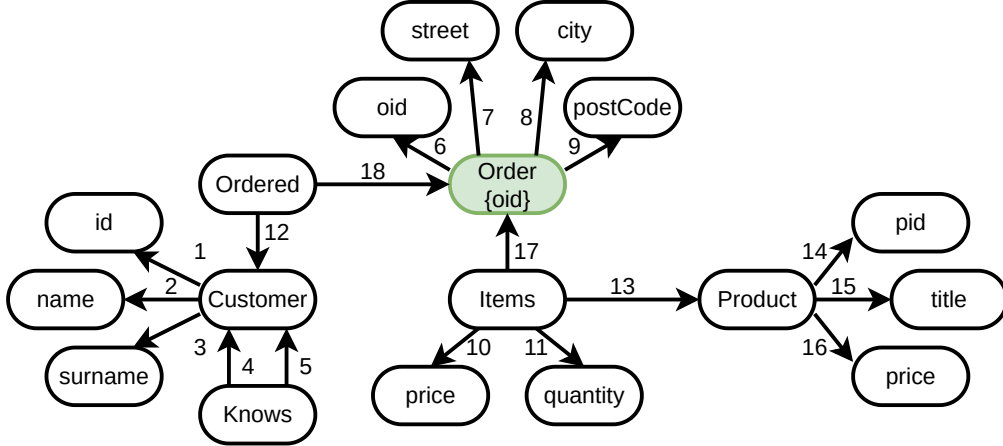




Order

```
{ _id : 2023001,
  customer: { id: 1, name: Mary, surname: Smith, knows: [
    { id: 2, name: John, surname: Newlin},
    { id: 3, name: Anne, surname: Maxwell}
  ]},
  street: Ke Karlovu, city: Prague, postCode: 110 00,
  items: [
    { pid: P5, title: Sourcery, quantity: 1, price: 350, currentPrice: 350 },
    { pid: P7, title: Pyramids, quantity: 1, price: 250, currentPrice: 275 }
  ]}
```

```
{ _id : 2023002,
  customer: { id: 2, name: John, surname: Newlin},
  street: Technická, city: Prague, postCode: 162 00,
  items: [
    { pid: P7, title: Pyramids, quantity: 1, price: 275, currentPrice: 275 }
  ]}
```

```
SELECT {
  _:knows friend ?friendA ;
    friend ?friendB .
  ?friendA fName ?friendAFName ;
    lName ?friendALName .
  ?friendB fName ?friendBFName ;
    lName ?friendBLName .
} WHERE {
  ?orders -18/12 ?friendA .
  ?orders -18/12 ?friendB .
  ?friendA -4/5 ?friendB .
  ?friendA 2 ?friendAFName ;
    3 ?friendALName .
  ?friendB 2 ?friendBFName ;
    3 ?friendBLName .
  FILTER(?friendA < ?friendB)
}
db.orders.find({ "customer.id":
  { $lt: "customer.knows.id" }
}, {
  "_id": 0,
  "customer.name": 1,
  "customer.surname": 1,
  "customer.knows.name": 1,
  "customer.knows.surname": 1
});
```

q<sub>1</sub>

```
SELECT {
  SUM(?price) AS ?totalPrice .
} WHERE {
  ?orders -18/12 ?customer ;
    -17/10 ?price .
  ?customer 2 "Anna" .
}
ORDER BY (?totalPrice) DESC
LIMIT

db.orders.aggregate([
  { $match: { "customer.name": "Anna" } },
  { $unwind: "$items" },
  { $group: {
    _id: "$_id",
    totalPrice: { $sum: "$items.price" },
  } },
  { $project: { _id: 0, totalPrice: 1 } },
  { $sort: { totalPrice: -1 } },
  { $limit: 1 }
]);
```

q<sub>2</sub>